

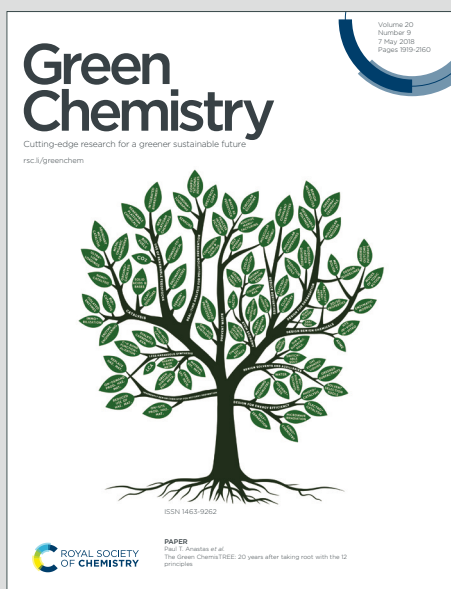
Green Chemistry

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Aqueous electrosynthesis of silver indium selenide nanocrystals and photothermal properties

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The eco-friendly aqueous electrosynthesis of AgIn₅Se₈ (AlSe) nanocrystals (NCs), and the respective core/shell structure AgIn₅Se₈/ZnS NCs (AlSe/ZnS), was carried out by using L-glutathione (GSH) as stabilizer. Se⁰ direct reduction was carried out in a graphite powder macroelectrode of an electrochemical cavity cell; controlled by current (*i* = 30 mA) and electrical charge (*Q* = 11.6 C). The generated Se²⁻ ions reacts with In³⁺ and Ag⁺ ions in the intermediary compartment of the cell to give AlSe NCs. A paired electrolysis (Zn⁰ anode/S⁰ in graphite powder cathode) was carried out to prepare ZnS-GSH NCs, using the same electrochemical apparatus (*i* = 30 mA, *Q* = 9.6 C). The effect of the Ag⁺/In³⁺ ratio (0.5, 0.28 and 0.18) and ZnS shell over AlSe NCs optical properties was investigated. The ternary AlSe NCs were characterized by XRD and HRTEM as a AgIn₅Se₈ cubic phase structure (Ag⁺/In³⁺ = 0.18); the ZnS shell was a zinc blende structure. The shell growth promoted a heterojunction formation with different crystalline phases, leading to a quantum yield decrease (QY: AlSe = 18.6 %; AlSe/ZnS = 16.0 %). The NCs were irradiated with near-infrared laser at different power densities (λ_{exc} = 785 nm). AlSe and AlSe/ZnS samples showed photothermic response, reaching 381 K and 402 K, respectively.

Introduction

The current socio-economic demand for eco-friendly materials with prospective applications on bioimaging and biosensing has increased in the last years especially for the case of non-invasive treatments.¹ Thus, the research for the development of quantum dots (QDs) structurally free of Cd, Pb and Hg has intensified in order to their high toxicity and teratogenic effects.² In the opposite way of the interface stability of nanocrystalline semiconductor systems A^{II}-C^{VI}, the ternary semiconductors A^I-B^{III}-C^{VI},² offer optical, structural and physico-chemical stability properties, making them promising candidates to bioapplications.³

The ternary nanocrystalline semiconductors, in particular, the AgIn₅Se₈ system, presents high potential for cell imaging applications when compared to Cd-based nanocrystals (NCs) and organic fluorophores. It occurs due to its photoluminescence (PL) observed in the near-infrared (NIR) to red region,⁴ good quantum yields (QY)⁵ and wide Stokes shift. These properties increase the resolution and reliability in the

imaging of biological tissues.⁶ Such characteristics are also interesting for the development of optical-electronic devices.⁷

The development of photothermal agents (PTAs) with satisfactory photothermic conversion and biocompatibility remains a research challenge.⁸ Some works describes the use of carbon dots,⁹ gold nanoparticles,¹⁰ and CuS QDs.¹¹ In this last class of nanoparticles, the semiconductor photothermal response of the AgIn₅Se₈ system still is unexplored.

New synthetic strategies in aqueous medium with high QY still offers a wide field of research.¹² Traditional methods for the synthesis of AgInSe₂ NCs generally involve expensive and toxic organic reagents, normally insoluble in aqueous medium and requires fine control of experimental conditions.¹³ Microwave,¹⁴ hydrothermal^{15,16} and hot injection¹² are among the well established aqueous medium synthetic routes. However, these strategies use the extrapolation of the water phase diagram, for supercritical conditions, where the dielectric constant is smaller, also being influenced by the operator's expertise.¹⁷

Electrosynthesis is a centenary methodology¹⁸ that can be applied to the preparation of both organic^{19,20} and inorganic²¹ compounds. At the forefront of the development of cleaner synthetic procedures and green chemistry evolution, the electrochemistry has been recognized as a promising methodology of synthesis, since it meets 9 of the 12 postulates of the green chemistry:^{22,23} 1) the use of green solvents like microemulsions, ionic liquids and water; 2) mediators and cat-

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alysts; 3) direct, indirect and paired electrolysis are in agreement with the concept of atom economy and avoid the use of toxic substances; 4) the use of renewable starting materials; 5) the energetic aspect is impacted when reactions are conducted at room temperature, or using mediators as electron carriers, or paired electrolysis are conducted; 7) real-time monitoring can be achieved during the electrolysis; 8) safety is improved and the chance of an accident is reduced by in situ electro-generating or recycling reactive species; 9) waste production can be prevented when reagents are stoichiometrically electrogenerated. Thus, when compared to traditional methods of synthesis, in aqueous or organic media, the electrochemical method offers additional advantages for the synthesis of semiconductor nanocrystals.

The electrochemical generation of chalcogen acids (H_2S , H_2Se and H_2Te), as reagent precursors, was the first electrochemical alternative used for the preparation of binary cadmium QDs in aqueous media.^{24–26} Ge²⁷ and Ham²⁸ described the preparation of electrodes of tellurium and selenium to electrochemically produce Te^{2-} and Se^{2-} , which were used in the synthesis of CdTe and CdSe QDs, furnishing nanocrystal precursors with a wide size dispersion.

Following the green chemistry concept, highly luminescent CdTe and CdSe QDs were produced by a clean and controllable method. Te^{2-} and Se^{2-} ions were electrochemically generated in basic (pH 14) aqueous medium and argon atmosphere, using a standard electrochemical cell with separated electrode compartments.^{29,30}

More recently, the electrochemical method for the generation of chalcogenide ions ($\text{X}^{2-} = \text{S}^{2-}$, Se^{2-} and Te^{2-}) was improved³¹ by the development of an electrochemical cavity cell,^{32,33} basically composed by a graphite powder macroelectrode and an auxiliary compartment containing the electrolyte solution/precursor metal ions. The elemental chalcogen is reduced into the macroelectrode, giving chalcogenide ions that are expelled to the aqueous electrolyte solution (cathode electrostatic repulsion). The nanocrystals are formed in a controlled way in the presence of a stabilizer agent, to furnish MX or M_2X_3 QDs.^{31,34,35}

The cavity cell also allows a paired electrosynthesis, where the QD precursor ions are simultaneously generated by reduction of the elemental chalcogen and oxidation of a metal sacrificial electrode.³¹ The cavity cell provides a particle nucleation control with minor operator interference (systematic error) and higher reproducibility. Moreover, the electrochemical reactions can be carried out in a simple and

low-cost experimental apparatus, soft and green, synthetic conditions.^{23,31,34–36} DOI: 10.1039/C9GC03647D

In this work, ternary semiconductor nanocrystals were electrosynthesized by the first time. The *in situ* preparation of Se^{2-} ions was carried out in a cavity cell and in presence of $\text{Ag}^+/\text{In}^{3+}$ ions, giving the Ag-In-Se ternary system (AISe). The L-glutathione (GSH) was used as stabilizer, a natural tripeptide, to improve the biocompatibility of the AISe NCs. A core/shell AISe/ZnS was also prepared in cavity cell, applying the paired electrolysis method. The performance of these nanomaterials was evaluated to photothermic applications.

Results and discussion

Structure and morphology of the electrochemically synthesized AISe and AISe/ZnS NCs

The optical properties of the AISe and AISe/ZnS systems were evaluated by using $\text{Ag}^+/\text{In}^{3+}$ ratios (0.5, 0.28, and 0.18) that belongs to the AISe cubic phase domain.³ Fig. 1A shows the powder X-ray diffraction patterns for the selected $\text{Ag}^+/\text{In}^{3+}$ ratios and ZnS shell, presenting the advance in crystalline phases. For all AISe NCs, it was observed the 2 θ diffraction angles at 27.5° and 45.5°, which are characteristic of planes (311), (203). Only the AISe/ZnS NCs, prepared in solution with $\text{Ag}^+/\text{In}^{3+}$ ratio = 0.18, presented the third 2 θ diffraction angle at 54°, characteristic of the plane (620), in agreement to the cubic structure of AgIn_5Se_8 (JCPDS 00-026-1477).^{3,37} For AISe/ZnS samples, it was observed the broadening of the peaks and shift to higher 2 θ values (28.5° and 47°) relative to ZnS coating, which is attributed to the decrease in lattice parameters due to the smaller ionic radius of Zn^{2+} relative to mono- and trivalent cations.³⁸ Thus, indicating the formation of the AISe/ZnS heterostructure, with a core/shell architecture.

The AISe and AISe/ZnS NCs ($\text{Ag}^+/\text{In}^{3+} = 0.28$) were also characterized by high-resolution transmission electron microscopy (HRTEM) equipped with an energy-dispersive X-ray analyzer (EDX) and selected area electron diffraction (SAED). Fig. 1B shows the HRTEM of ternary AISe NCs. It can be observed oval-shaped nanocrystals and assigned an interplanar distance of 0.32 nm for the (311) plane of the AISe cubic structure. In addition, the EDX spectrum (Fig. 1C) shows all the spectroscopic signals stated to AISe elements: Ag $L_{\alpha 1}$ (2.983 eV), In $L_{\alpha 1}$ (3.286 eV) and $L_{\beta 1}$ (3.487 eV), Se $L_{\beta 1}$ (1.419 eV) and $K_{\alpha 1}$ (11.224 eV). The additional signal of sulphur, S $K_{\alpha 1}$ (2.309 eV), is due to the thiol group of the GSH stabilizer. Fig. 1F presents the HRTEM of AISe/ZnS-GSH NCs, where the

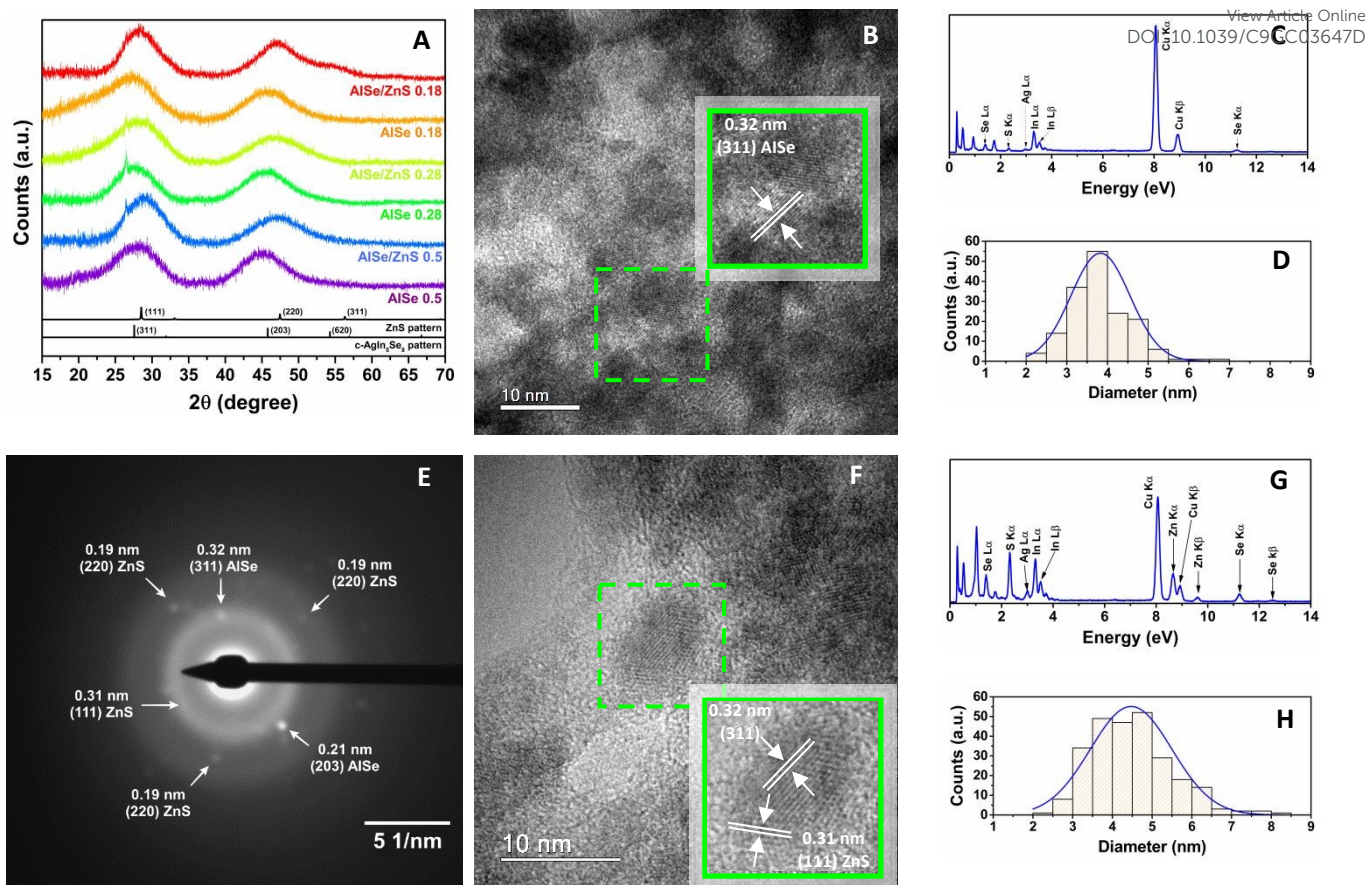


Fig. 1 Structural characterization of AlSe and AlSe/ZnS NCs. (A) XRD, crystalline phases of AlSe and AlSe/ZnS NCs ($\text{Ag}^+/\text{In}^{3+} = 0.5, 0.28$ and 0.18). (B) HRTEM images, (C) EDX spectrum and (D) corresponding crystallite size distribution histograms of AlSe NCs ($\text{Ag}^+/\text{In}^{3+} = 0.28$). (E) Selected area electron diffraction (SAED) for AlSe/ZnS ($\text{Ag}^+/\text{In}^{3+}$ ratio = 0.28). (F) HRTEM images, (G) EDX spectrum and (H) corresponding crystallite size distribution histograms of AlSe/ZnS NCs ($\text{Ag}^+/\text{In}^{3+} = 0.28$).

difference on the AlSe and ZnS heterostructures can be observed by contrast. The core presents the planes (311) of the AlSe and the shell presents the interplanar distances of 0.31 nm, characteristic of the zinc blend plane (111).³⁹ The EDX analysis of the AlSe/ZnS nanoparticles (Fig. 1G) allowed the identification of Ag, In and Se elements, as observed in Fig. 1C. $\text{Zn } k_{\alpha 1}$ (8.630 eV) and a more intense peak of $\text{S } k_{\alpha 1}$ (2.309 eV) were additionally identified, corresponding to the ZnS shell composition. The average size (Fig. 1D) of the AlSe NCs (3.8 ± 0.7 nm) increased to 4.5 ± 1.0 nm, after the coverage with ZnS (Fig. 1H), which is indicative of the core coating. Fig. 1E shows the SAED pattern for the AlSe/ZnS nanocrystal. Diffuse rings show that the nanocrystal is polycrystalline. The selected diffraction rings and atomic positions present the cubic planes of AlSe: (311) and (203); and ZnS: (111) and (220).

The dynamic light scattering (DLS) measurements showed that the hydrodynamic radius (H_R) of the AlSe NCs stabilized by GSH decreases with the $\text{Ag}^+/\text{In}^{3+}$ ratio diminishing (Table 1, entries 1 to 3). Thus, the increase of nanoparticle radius can be associated to the increase of Ag^+ concentration in the nanocrystal lattice, due to its larger ionic radius ($\text{Ag}^+ = 115$ pm) if compared to In^{3+} ion (80 pm). And also can be associated to the nucleation mechanism and nanoparticle growth that can increase the number of defects (voids) into the crystal lattice.⁴⁰ For the AlSe/ZnS NCs stabilized by GSH, it

was observed the same behavior that AlSe NCs (Table 1, entries 4 to 6), related to the $\text{Ag}^+/\text{In}^{3+}$ ratio increase. In addition, the core/shell H_D showed larger values when compared to the core, which is expected because additional ZnS shell, promoting an additional volume to the nanoparticle, as seen in the HRTEM images (Fig. 1).

The nanoparticle negative charge surface was confirmed for all AlSe and AlSe/ZnS NCs, by zeta potential (ZP) assays (Table 1). At pH 9, the GSH carboxylate and thiol groups are deprotonated, and the nanoparticle-S-GSH interaction should be favored. The AlSe and AlSe/ZnS ZP values (*i.e.*, $|\pm\text{ZP}|$), are inversely proportional to the $\text{Ag}^+/\text{In}^{3+}$ ratio (Table 1, entries 1 to 6). It occurs because the increase of the nanoparticle surface area, as observed in H_D measurements, while the GSH concentration for all AlSe and AlSe/ZnS solutions remains constant, thus, the charge number (GSH molecules/nanoparticle surface) diminishes. The ZP values determined for the AlSe and AlSe/ZnS nanocrystals indicate good stability.

Infrared spectra (Fig. S1, Supplementary Information) evidence the interaction between the deprotonated GSH stabilizer and the synthesized nanoparticles, acting as a stabilizer and its modes of coordination. The spectra were recorded at pH 9 for GSH, AlSe-GSH and AlSe/ZnS ($\text{Ag}^+/\text{In}^{3+}$ ratio = 0.28). For all samples, the peak at 2524 cm^{-1} relative to

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the S-H group was absent, indicating deprotonation and coordination to the AlSe and AlSe/ZnS surface. The stretching of the -NHR group at 1584 cm⁻¹ is present in the GSH sample, but the peak intensity decreases in the presence of nanocrystals, indicating another mode of coordination.^{41–43}

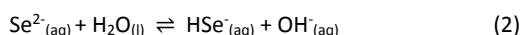
Table 1. Hydrodynamic radius (H_R) and zeta potential (ZP) parameters of the of AlSe and AlSe/ZnS NCs, according to Ag⁺/In³⁺ ratio used in the AlSe synthetic procedure.

Entry	System	Ag ⁺ /In ³⁺	H_R	ZP
1	AlSe	0.50	76.6 ± 7.4	-22.5 ± 1.6
2		0.28	38.8 ± 3.8	-33.9 ± 2.9
3		0.18	25.2 ± 2.7	-35.8 ± 4.6
4	AlSe/ZnS	0.50	113.5 ± 9.1	-18.8 ± 2.6
5		0.28	52.1 ± 7.4	-21.1 ± 2.9
6		0.18	33.1 ± 2.2	-23.1 ± 2.6

Electrochemical preparation of AlSe and AlSe/ZnS NCs

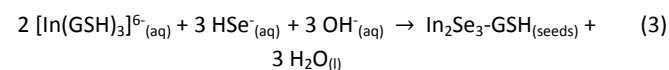
The application of the electrochemistry on the synthesis of AlSe and AlSe/ZnS NCs is focused on obtaining the precursor ions Se²⁻ and S²⁻. Electrochemistry makes possible the generation of these ions *in situ*, without the need of chemical reducing agents.^{29–31,34,35} The generation of by-products or the need for counter-ions is avoided, making the electrochemical method cleaner. In the case of metallic precursors, the respective silver and indium salts are used due to the difficulty of simultaneous oxidation of Ag⁰ and In⁰, during the electrolysis process. However, in the case of ZnS shell preparation, Zn²⁺ ion can be simultaneously produced with S²⁻ ions through a paired electrolysis.³⁴

The growth mechanism of the silver and indium selenide NCs was initiated by a coordination chemistry step, where the Ag⁺ and In³⁺ ions were coordinated to the deprotonated GSH (pH = 9, aqueous solution), giving the respective [Ag(GSH)₂]⁵⁻ and [In(GSH)₃]⁶⁻ complexes,^{44–46} which were stored into the intermediate compartment of the electrochemical cavity cell (Fig. S2A, Supplementary Information). Different Ag⁺/In³⁺ ratios were tested (0.5, 0.28, 0.18) to observe the evolution of optical and structural properties for the selected AlSe cubic phase. Into the graphite powder macroelectrode, Se⁰ is reduced to Se²⁻ (Eq. 1), which is simultaneously expelled (electrostatic repulsion) to the intermediate compartment of the cavity cell. At pH 9, HSe⁻_(aq) is formed (Eq. 2) in agreement to the selenium Pourbaix diagram.⁴⁷ The control of the reduction process, with a constant current of 30 mA and charge (Q) = 11.6 C, guarantees the formation of the species in a constant and uniform way.

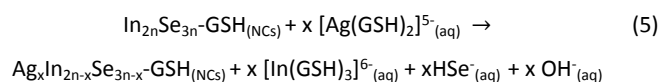


At this point, the AlSe formation is mediated by the solubility product of the species in solution (Ag₂Se, $K_{sp} = 1.8 \times 10^{-54}$; In₂Se₃, $K_{sp} = 5.6 \times 10^{-92}$),^{48,49} and by the balance between the coulombic contributions and the molecular interactions (frontier molecular orbitals), according to Pearson-Klopman acids (Ag⁺ e In³⁺) and bases (Se²⁻).^{50,51} The first parameter is

related to the ionic character of interaction, and the second is relative to the covalent interaction among ions. As Se²⁻ is a soft base and Ag⁺ is a soft acid, the Ag⁺/Se²⁻ interaction should be favoured when compared to the Se²⁻ and In³⁺ (hard acid) interaction. The former is more representative of a covalent interaction. However, the solubility of Ag⁺ ([Ag⁺] = 1.8×10^{-18} mol.L⁻¹) is one order of magnitude higher than In³⁺ ([Ag⁺] = 4.4×10^{-19} mol.L⁻¹) for selenide salt.⁵² Therefore, the formation of In₂Se₃-GSH is initially favoured, and the nucleation step is presented in Eqs. 3 and 4.



After the formation of the In₂Se₃-GSH seeds and crystal growth, a ternary nanocrystal is generated by solid-state cation-exchange, via Frenkel defects (Eq. 5). The Ag⁺ assumes the position of In³⁺ cations, leading to formation of Ag_xIn_{2n-x}Se_{3n-x}-GSH NCs, re-introducing In³⁺ and Se²⁻ ions in the solution that can generate new In₂Se₃-GSH seeds.^{5,49,53}



The Ag⁺/In³⁺ cation-exchange into the nanocrystal lattice can be controlled by Ag⁺/In³⁺ ratio in solution, allowing the crystalline phase modification, as observed in the diffractograms shown in Fig. 1A. Beyond the discrete phase changing, the excess of silver promotes the insertion of new intragap states and tuning of optical band-gap.^{48,53,54} Thus, the Ag_xIn_{2n-x}Se_{3n-x}-GSH generic structure proposed in Eq. 5 can be associated to possible crystalline phases of AlSe NCs: AgInSe₂ (n = 1, x = 1), AgIn₃Se₅ (n = 2, x = 1) or AgIn₅Se₈ (n = 3, x = 1).

Fig. 2A shows the UV-Vis-NIR spectra recorded during the electrolysis step, it can be observed the increase of the baseline absorption, which is related to In₂Se₃-GSH nucleation process. The heat treatment (100°C) of the post-electrolysis solution showed the same optical behaviour of classical quantum dots, *i.e.*, the quantum confinement associated to the nanoparticle size. After 5 min of heating, it was observed the appearing of an absorption band at $\lambda_{\text{max}} = 525$ nm (Fig. 2B), followed by a shift to 530 nm after 30 min, which remained unaltered for greater heating time. This is a typical behaviour of systems with many intragap states, promoting several reabsorption processes.⁵⁵

The band energy of the AlSe and AlSe/ZnS nanocrystals was determined by analyzing the Tauc plots ($(\alpha h\nu)^{1/r}$ vs $h\nu$), where α is the absorption coefficient, h is the Planck constant and ν is the light frequency. The r value of the exponent defines the electronic nature of nanocrystals and can be presented as 3, 2, 3/2 and 1/2, corresponding to indirect forbidden, indirect allowed, direct forbidden and direct allowed transitions, respectively.⁵⁶ AlSe QDs have direct allowed electronic transitions, so $r = 1/2$.⁵⁷

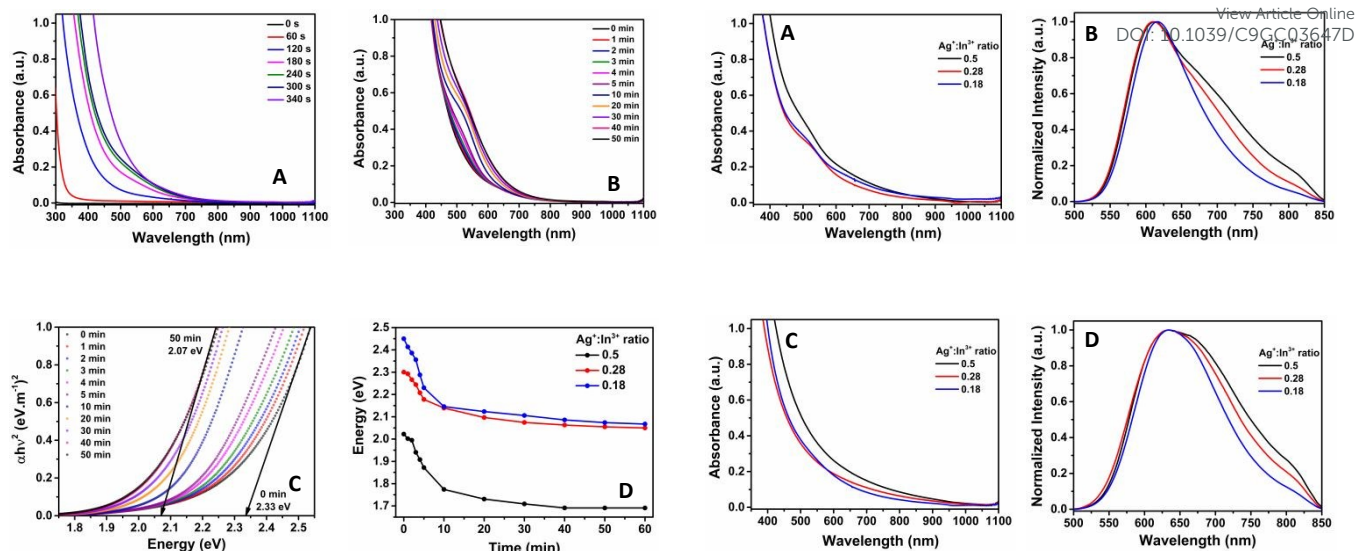


Fig. 2. (A) UV-Vis-NIR absorption spectra for AISE ($\text{Ag}^+/\text{In}^{3+} = 0.28$) seeding evolution during the electrolysis. (B) Absorption spectra and (C) Tauc plot for AISE nanocrystals ($\text{Ag}^+/\text{In}^{3+} = 0.28$) at different heating (100°C) times. (D) Optical band gap vs heat treatment time for AISE ($\text{Ag}^+/\text{In}^{3+}$ ratios = 0.5, 0.28 and 0.18).

Fig. 2C shows the Tauc plots of the respective AISE-GSH UV-Vis-NIR spectra (Fig. 2B). The optical bandgap of the AISE-GSH NCs ($\text{Ag}^+/\text{In}^{3+}$ ratio = 0.28) ranged from 2.33 eV to 2.07 eV, for 0 min to 50 min of heat treatment, respectively. Fig. 2D shows the evolution of band gap value vs heating time for AISE NCs prepared at $\text{Ag}^+/\text{In}^{3+}$ ratio = 0.5, 0.28 and 0.18. It can be observed a lower band gap for nanocrystals prepared at $\text{Ag}^+/\text{In}^{3+} = 0.5$ (1.7 eV after 30 min), while the nanocrystals prepared at $\text{Ag}^+/\text{In}^{3+}$ ratios = 0.28 and 0.18 showed similar results with higher band gap values at 2.08 eV and 2.1 eV, respectively.

UV-Vis-NIR absorption and emission profiles of AISE-GSH NCs are shown in Figs. 3A and 3B, respectively. As already described, the decreasing of the $\text{Ag}^+/\text{In}^{3+}$ ratio leads to the slight increase of the optical band gap, justified by smaller number of crystalline defects achieved in a purer AgIn_5Se_8 nanocrystal.³

The AISE-GSH emission spectra (Fig. 3B) show a band at $\lambda_{\text{max}} = 610$ nm, which is located between emission bands of In_2Se_3 -GSH ($\lambda_{\text{max}} = 433$ nm) and Ag_2Se -GSH ($\lambda_{\text{max}} = 830$ nm) NCs, prepared by the same electrochemical procedure (cavity cell), Fig. 4A. The decreasing of the $\text{Ag}^+/\text{In}^{3+}$ ratio (in solution) leads to the narrowing of the emission bands (FWHM are described in Table 2), without significant displacement of the maximum emission wavelength. Only the $\text{Ag}^+/\text{In}^{3+}$ ratio = 0.18 presented the Gaussian shape, while the 0.28 and 0.50 samples presented asymmetric and broad emission bands. Thus, it is expected that the 0.18 ratio correspond to an ideal $\text{Ag}^+/\text{In}^{3+}$ concentrations to obtain nanocrystals with the AgIn_5Se_8 crystalline phase, in agreement to XRD characterization (Fig. 1A). While the AISE colloidal solutions obtained at 0.28 and 0.50 ratios should contain a mixture of AgInSe_2 , AgIn_3Se_5 and AgIn_5Se_8 (majority).

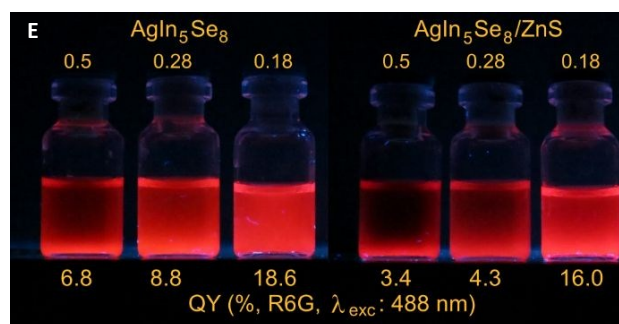


Fig. 3 (A) UV-Vis-NIR and (C) PL spectra of AISE nanocrystals. (B) UV-Vis-NIR and (D) PL spectra of AISE/ZnS nanocrystals. (E) Image of AISE and AISE/ZnS samples prepared at different $\text{Ag}^+/\text{In}^{3+}$ ratios (0.5, 0.28 and 0.18) under UV light = 365 nm, and the respective QYs determined under $\lambda_{\text{exc}} = 488$ nm.

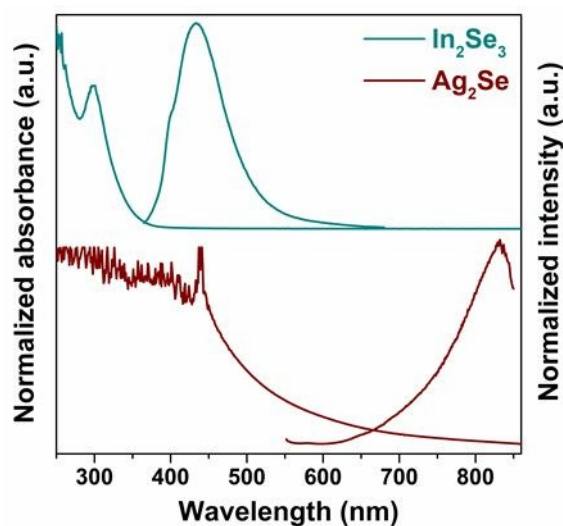


Fig. 4 (A) UV-Vis-NIR absorption and emission spectra of In_2Se_3 -GSH and Ag_2Se -GSH NCs. Inset: In_2Se_3 -GSH (left) and Ag_2Se -GSH (right) samples under UV light, $\lambda_{\text{exc}} = 365$ nm.

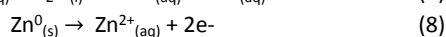
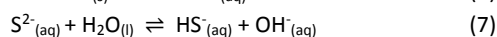
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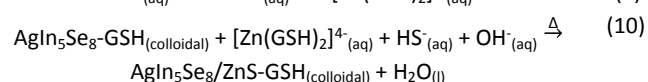
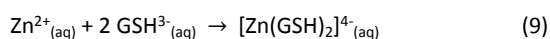
Table 2. Optical properties of AISE and AISE/ZnS systems, prepared at different $\text{Ag}^+/\text{In}^{3+}$ ratios (0.5, 0.28 and 0.18): Band gap energy (E_g), maximum wavelength emission (λ_{em}), full width at half maximum (FWHM), and quantum yield (QY).

Entry	Sample ($\text{Ag}^+/\text{In}^{3+}$)	E_g (eV)	λ_{em} (nm)	FWHM (nm)	QY (%)
1	AISE (0.50)	1.77	616	149	6.8
2	AISE (0.28)	2.13	612	133	8.8
3	AISE (0.18)	2.14	610	107	18.6
4	AISE/ZnS (0.50)	2.06	635	171	3.4
5	AISE/ZnS (0.28)	2.18	633	160	4.3
6	AISE/ZnS (0.18)	2.25	634	136	16.0

The electrochemical preparation of the AISE/ZnS heterojunction was carried out by coating the AISE with ZnS NCs prepared in a cavity cell (Fig. S2B) by an *in situ* paired electrochemical reaction: simultaneous reduction of S^0 (Eqs. 6 and 7) and oxidation of Zn^0 (Eq. 8).



In the presence of GSH and pH 9, the $[\text{Zn}(\text{GSH})_2]^{4-}$ complex is formed (Eq. 9), following the ZnS seeding by reaction with HS^- , simultaneously generated (Eq. 10).



The growth of ZnS shell on the AISE structure is also related to the surface chemical potential of the ternary nanocrystal.⁴⁹ The heterojunction AISE/ZnS should occur *via* epitaxial growth, with exchange of Ag^+ and In^{3+} by Zn^{2+} ions *via* Frenkel defects. However, the natural polarizability difference of the chalcogenides, S^{2-} and Se^{2-} , leads to a distortion of the parameters of the lattice, leading to different crystalline phases between core and shell (PXRD, Fig. 1A).^{58,59} Such behavior is fundamental to the processes of energy transfer and recombination, directing to a nanoparticle with an alloy or core/shell structure composition.^{49,60}

The heat treatment (under reflux, 100°C) implies to a smaller parameter of Pearson-Klopman interaction,⁵¹ since the dielectric constant decreases as a function of the heating, which favours the nanoparticle growth process. After the addition of ZnS shell to the AISE structure, it was observed the smoothing in UV-Vis absorption bands (Fig. 3C), making it a continuous band. The AISE/ZnS-GSH NCs emission spectra (Fig. 3D) showed the enlargement of the emission bands.^{3,5} The Fig. 3E shows the samples of AISE and AISE/ZnS NCs under UV excitation ($\lambda_{\text{exc}} = 365$ nm). All optical properties of AISE-GSH and AISE/ZnS-GSH were summarized in Table 2. The decrease of the $\text{Ag}^+/\text{In}^{3+}$ ratio led to an increase of the optical bandgap value and a reduction of the FWHM. The ZnS shell deposited on the AISE surface altered the optical profile of the AISE NCs, where the absorption bands (Fig. 3C) became continuous,

showing that the ZnS shell promotes the insertion of new intragap states at the semiconductor's interface. These experimental data reinforce the proposed mechanism of heterojunction between AISE and ZnS.

The QY ($\lambda_{\text{exc}} = 488$ nm) of the AISE nanocrystals (Table 2) increases with decrease of the $\text{Ag}^+/\text{In}^{3+}$ ratio ($\text{Ag}^+/\text{In}^{3+} = 0.5$: QY = 6.8%; 0.28: 8.8%; 0.18: 18.6%). Thus, the higher QY observed for AISE-GSH prepared at ratio = 0.18 shows a better condition to give AgIn_5Se_8 crystalline structure, decreasing the number of intragap states and recombination probability. By deposition of the ZnS shell on the AISE surface, it can be observed the QYs decrease for all $\text{Ag}^+/\text{In}^{3+}$ ratios ($\text{Ag}^+/\text{In}^{3+} = 0.5$: QY = 3.4%; 0.28: 4.3%; 0.18: 16.0%), which can be related to the new crystalline/electronic defects added by ZnS shell on the AISE surface.

The synthetic parameters and results reported in this work were compared to data of literature (Table S1, Supplementary information).^{3,12,16,62} Regarding the synthetic parameters, the electrochemical synthesis occurs at ambient temperature and atmospheric pressure, while the hydrothermal/solvothermal methods occur under higher temperature and pressure. In the present synthetic approach, the direct reduction of Se^0 or S^0 is controlled by constant current and electrical charge, and it was carried out without the use of toxic reducing reagents, which is aligned to the green chemistry principles. In addition, the slow introduction of Se^{2-} or S^{2-} ions in the reaction medium (controlled current) allows to control the nucleation step of AISE NCs, which is another differential of the electrochemical procedure.^{31,34}

Concerning the optical properties, the emission wavelengths of the AISE NCs prepared in aqueous medium ranged from 575 to 650 nm, Table S1, entries 3 to 6.^{12,16} The emission wavelength of NCs prepared by hot-injection method in organic medium (n-trioctylphosphine) varied from 700 to 850 nm (entry 1).^{3,62} The NCs prepared by the same method in 1-octadecene solvent (oleylamine/dodecanethiol stabilizer) presented emission wavelength at 619 nm (entry 2). The QY of for NCs prepared in organic solvents were 5.0 and 6.9% (entries 1 and 2, respectively). In aqueous medium, the NCs stabilized by GSH (QY = 10.3%, entry 3) and gelatin/MPA (QY = 5.0%, entry 4) promoted a satisfactory stability. In the other hand, NCs prepared by the electrochemical method and stabilized by GSH presented good QYs: AISE-GSH (18.6%) and AISE/ZnS-GSH (16.0%). Thus, the electrosynthesis of AISE and AISE/ZnS give NCs with satisfactory QY at ambient reaction conditions, following the green chemistry principles.

Photothermal effect measurements

For all samples of AISE NCs, the QY value decreased after the coating with ZnS. Thus, the ZnS shell implies to the increase of non-radiative decay processes,⁶³ leading to promising photothermic response and application on hyperthermia measurements. The AISE/ZnS NCs ($\text{Ag}^+/\text{In}^{3+}$ ratio = 0.28) was chosen because it presented the larger QY decrease, 4.5% (Table 2). The photothermal performance induced by near-infrared laser irradiation ($\lambda_{\text{exc}} = 788$ nm) was

investigated by using the AISE and AISE/ZnS powders, obtained from precipitation of the NCs by addition of acetone. It was used a power density ranging from 400 to 1000 mW.cm^{-2} , during 10 min. The thermal vs. time response curves are presented in Figs. 5A and 5B.

For all samples, the temperature increases linearly with the power density up to 800 mW.cm^{-2} , then a decrease of the slope is indicative of the beginning of a saturation process (Fig. 5C). The AISE/ZnS sample excited at 400 mW.cm^{-2} achieved 319 K, which is a sufficient temperature to induce the irreversible cell death.^{64,65} The AISE and AISE/ZnS NCs reached temperatures of 377 K and 400 K, respectively, when irradiated at 1000 mW.cm^{-2} . The photothermal performance of AISE and AISE/ZnS NCs reached values higher than those reported for CuFeS_2 (343 K, $\lambda_{\text{exc}} = 808 \text{ nm}/2000 \text{ mW.cm}^{-2}$, and CuS (325 K, $\lambda_{\text{exc}} = 980 \text{ nm}/72 \text{ mW.cm}^{-2}$).⁶⁶

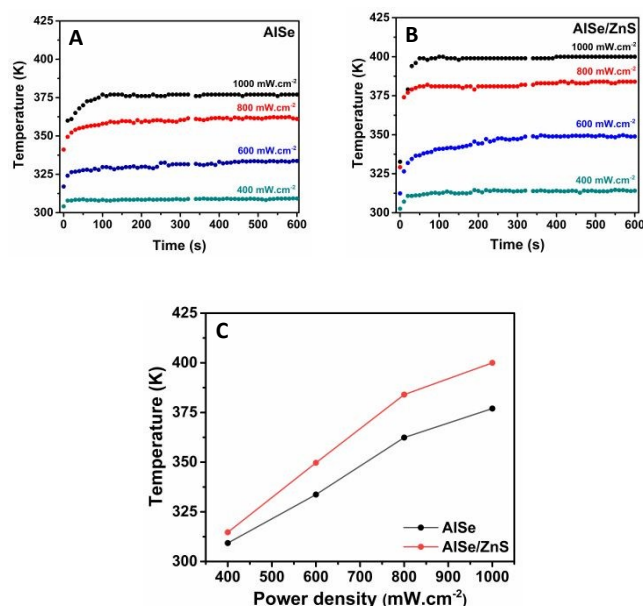


Fig. 5 Thermal vs. time response for (A) AISE and (B) AISE/ZnS NCs ($\text{Ag}^+/\text{In}^{3+} = 0.28$ and $\lambda_{\text{exc}} = 785 \text{ nm}$) at 400, 600, 800 and 1000 mW.cm^{-2} . (C) Thermal vs power density response for AISE and AISE/ZnS NCs ($\text{Ag}^+/\text{In}^{3+} = 0.28$ and $\lambda_{\text{exc}} = 785 \text{ nm}$).

Experimental

Materials

Silver (I) nitrate (AgNO_3 anhydrous, 99.99%), sodium perchlorate (NaClO_4 , $\geq 98\%$), elemental selenium powder (Se^0 , 99.99%, -100 mesh), elemental sulfur powder (S^0 , 99.98%, -100 mesh) and graphite powder, particle size < 20 μm) were purchased from Sigma-Aldrich; Indium (III) bromide trihydrate ($\text{InBr}_3 \cdot 3\text{H}_2\text{O}$, 99.99%) were purchased from Alfa-Aesar. The elemental zinc (Zn , 99.9%) bar was purchased from Goodfellow. Ethanol and acetone were P.A. grade and purchased from Dynamic. All reagents were used without prior purification. Aqueous solutions were prepared with deionized

Milli-Q water (18.0 $\Omega\text{.cm}$). All chemicals were used without pre-treatment. Stock solutions of AgNO_3 (0.25 mol.L^{-1}) and $\text{InBr}_3 \cdot 3\text{H}_2\text{O}$ (0.25 mol.L^{-1}) were prepared in water and ethanol, respectively. All stock solutions were stable for 1-2 weeks.

Electrosynthesis of the AISE NCs

The electrolysis procedure was carried out by using a potentiostat/galvanostat PGSTAT 30 Autolab (Metrohm) and Nova 2.1 software. A detailed description of the electrochemical cavity cell (Fig. S2A) assembly used for the AISE electrosynthesis was previously described by our group.^{31,34}

The cathodic compartment of the cavity cell (graphite powder macroelectrode) was filled with 71.8 mg (6.0 mmol) of graphite powder mixed to 4.73 mg (0.06 mmol) of elemental selenium powder (Se^0), and the mixture was pressed under $P = 3.2 \text{ kg.cm}^{-2}$ for 10 min. A sintered glass (previously sonicated in 0.1 mol.L^{-1} NaClO_4 aqueous solution, and with the same cavity cell diameter = 1.0 cm), was placed over the cathodic cavity to avoid the dispersion of the graphite, allowing at the same time the migration of selenide ions (Se^{2-}), formed during the electroreduction.

0.25 mol.L^{-1} AgNO_3 aqueous solution (volume = 120, 68 or 43 μL) and 0.25 mol.L^{-1} InBr_3 ethanoic solution (240 μL) were used to prepare $\text{Ag}^+/\text{In}^{3+}$ precursor solutions (0.5, 0.28, and 0.18 ratios, respectively), and added to 25 mL volumetric balloon containing 9 mL of Milli-Q water. GSH stabilizer (0.25 mmol) dissolved in 10 mL of Milli-Q water was added to the $\text{Ag}^+/\text{In}^{3+}$ solution, followed by 5 mL of NaClO_4 (2 mmol) solution. The pH of the final solution was adjusted to 9 by the dropwise addition of 1.0 mol.L^{-1} NaOH solution, then adjusting the water level into the cell by the addition of water (Fig. S2). The $\text{Ag}^+/\text{In}^{3+}$ ratio was varied from 0.03 to 0.0108 mmol of Ag^+ , while the In^{3+} amount was fixed to 0.06 mmol.

A stainless steel grid was used as an anode, placed in the anodic compartment containing 0.1 mol.L^{-1} NaOH solution, and separated from the intermediate compartment by a Nafion® membrane. Electrolysis was carried out at 30 mA constant current for a period of 386 s ($Q = 11.6 \text{ C}$), under argon. The AISE NCs formation can be identified by the solution colour changing in the intermediate compartment, which becomes brown-yellow that presents luminescence. The AISE seeds prepared under different $\text{Ag}^+/\text{In}^{3+}$ ratios were heated for 60 minutes under reflux and stored at 4°C in the refrigerator.

Electrosynthesis of Ag_2Se and In_2Se_3 NCs

The preparation of GSH stabilized Ag_2Se and In_2Se_3 nanocrystals followed the same electrochemical procedure reported for AISE NCs. Using the Me/Se^0 ratio, ($\text{Ag}/\text{Se} = 2:1$ and $\text{In}/\text{Se} = 2:3$). A graphite powder macroelectrode was prepared with 0.06 mmol Se^0 and introduced into the cavity cell. The solutions of Ag^+ (0.12 mmol), or In^{3+} (0.04 mmol), + 0.25 mmol GSH stabilizer + 2.0 mmol NaClO_4 electrolyte were prepared in 25 mL water MilliQ total volume, the pH was adjusted to 9, and the final solution was added to the central

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compartment of the cell. Electrolyses were conducted at 30 mA constant current for 321 s ($Q = 9.6$ C) under argon. The seeds of Ag_2Se or In_2Se_3 were heated for 60 min under reflux, and conditioned at 4°C in a refrigerator.

Electrochemical *in situ* preparation of ZnS for AlSe/ZnS growth

The preparation of AlSe/ZnS nanoparticles were also carried out in the electrochemical cavity cell (Fig. S2B), using a Zn^0 (rod) sacrificial anode. The preparation of the cathodic compartment occurred similarly to that described for the AlSe preparation, using 0.05 mmol of elemental sulfur mixed to 5.0 mmol of graphite powder. The solution 0.25 mmol GSH stabilizer + 2.0 mmol NaClO_4 electrolyte was prepared in 25 mL water MilliQ, the pH was adjusted to 9, and the final solution was added to the central compartment of the cell. The paired electrolysis was carried out at 30 mA constant current for 321 s ($Q = 9.6$ C). The AlSe/ZnS NCs prepared were heated under reflux for 30 min. The samples were kept at 4°C in a refrigerator.

Photothermal effect measurements

FLIR E5 infrared camera was used to measure the photothermal effect data. A 30 mg of AlSe and AlSe/ZnS was put into a quartz slide, then it was irradiated with an Ocean Optics near-infrared laser ($\lambda_{\text{exc}} = 785$ nm) for 10 min, the power density was varied between 400 mW.cm^{-2} to 1.000 mW.cm^{-2} . The uncovered slide was used as a negative control. The temperature was recorded at 10 s intervals by using a digital camera.

Characterization of AlSe nanocrystals.

Absorption spectra were acquired in quartz cuvettes using an Agilent 8453 UV-Vis spectrophotometer, and the Tauc curves were used to determine the optical bandgap. The emission spectra were obtained in the Fluorog Horiba Jobin Yvon spectrofluorimeter (with excitation and emission slit 3 mm) with $\lambda_{\text{exc}} = 488$ nm. The photoluminescence QY was determined at room temperature by comparing the integrated emission intensity of AlSe and AlSe/ZnS NCs in solution, and compared to the rhodamine 6G (R6G) standard in anhydrous ethanol, QY = 95%, with identical optical densities, $\lambda_{\text{exc}} = 488$ nm.⁶⁷

The morphological and structural characterization of the AlSe and AlSe/ZnS nanoparticles was based on HRTEM images, X-ray spectra (XRS), energy dispersive X-ray spectrometry (EDS) and selected area electron diffraction (SAED). A Tecnai G²-20-FEI high-resolution transmission electron microscope (HRTEM) was used (FEI Company, USA), with 200 kV acceleration potential, using copper-coated carbon screens (Electron Microscopy Sciences, USA) as support for the nanoparticles, followed by complete drying at low pressure overnight. The size distribution of the nanoparticles was obtained based on HRTEM images, randomly selecting 200 nanoparticles, using the image processing program (ImageJ, version 1.50, public domain, National Institutes of Health). The crystalline structure of the nanocrystals was obtained using

powder X-ray diffraction (PXRD) (Rigaku) equipped with X-ray tube (Cu K_α : $1.540 \text{ \AA}$, 40 kV, 40 mA, 2θ range from 10° to 80° , and step of 0.03°). The nanocrystals were precipitated using a 1:1 ratio of acetone/colloidal solution, submitted to centrifugation (30 min, 6000 rpm) and washed with ethanol. The samples were dried under vacuum, macerated in an agate mortar, and the powder was stored under inert atmosphere.

The size of the particles in the colloidal state was evaluated by dynamic scattering of light (DLS) and Zeta potential (ZP), using ZetaPlus equipment (Brookhaven Instruments Corporation, 35 mW red laser light, wavelength $\lambda = 660$ nm) from 25 points. ZP measurements were performed at $25 \pm 2^\circ\text{C}$ by the Smoluchowski approximation method in a set of 25 replicates. Diffuse reflectance infrared Fourier transform spectroscopy spectra (DRIFTS) were obtained in a Thermo Fischer Spectrometer (Nicolet 6700) over the range of $400 - 4000 \text{ cm}^{-1}$, using 64 scans and a 2 cm^{-1} resolution.

Conclusions

The electrochemical procedure described in this work showed reproducibility and versatility, making possible the preparation of binary, ternary and core/shell nanocrystals in aqueous medium. And efficacy, when a paired electrolysis was used for the preparation of ZnS NCs, using only electrons to generate the reaction precursors (Zn^{2+} and S^{2-}). The controlled addition of chalcogenide ions in the reaction medium can be considered a differential propertie of the electrosynthesis process.

AlSe and AlSe/ZnS NCs were synthesized by a green electrochemical methodology with crystalline phase control, giving fluorescence QYs up to 18.6%. The nucleation step can be controlled by the *in situ* electrochemical generation of Se^{2-} , giving In_2Se_3 seeds and gradual Ag^+ cation-exchange into the nanocrystal, furnishing mostly the AgIn_5Se_8 cubic crystalline phase. The *in situ* electrochemical preparation of ZnS allowed the formation of core/shell AlSe/ZnS nanoparticles with a heterojunction of different crystalline structures. Under $\lambda_{\text{exc}} = 785$ nm the nanocrystals showed photothermal activity, reaching temperature variations between 377 and 400 K, when irradiated at power density of 1000 mW.cm^{-2} . Thus, AlSe and AlSe/ZnS NCs showed potential hyperthermic applications.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

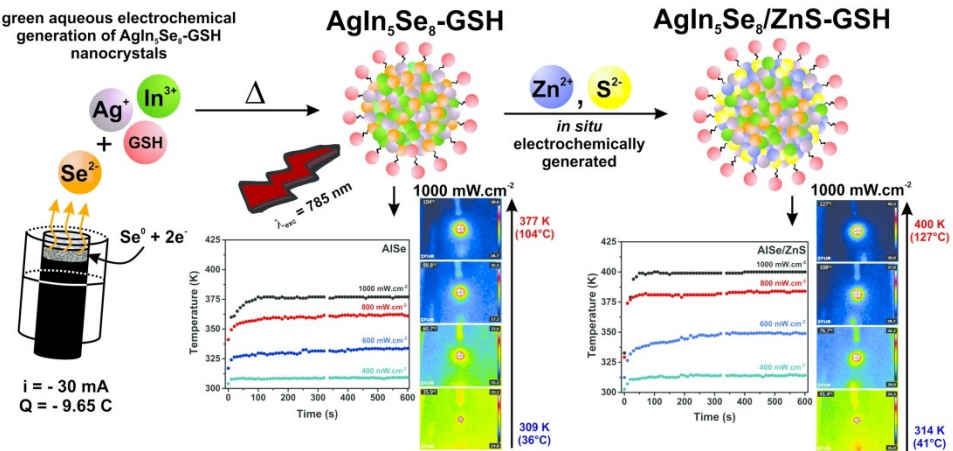
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